

A Machine Learning Approach to Volatility Forecasting

Replication and critical discussion of Christensen, Siggaard & Veliyev (2023, *Journal of Financial Econometrics* 21(5)) — AAPL, AMZN, JPM 5-minute data, 2016–2024.

1. The problem and the original paper

Realised variance (RV), the sum of squared intraday returns, is the ex-post measure of a day's return variation (Andersen & Bollerslev 1998) and the input to risk management, option pricing and allocation; unlike returns it is strongly predictable because volatility is persistent. The field benchmark is the Heterogeneous Autoregressive model (HAR; Corsi 2009): a linear regression of RV on its daily, weekly (5-day) and monthly (22-day) lagged averages — four parameters that approximate the long memory of volatility so well that decades of richer models struggle to beat it (Hansen & Lunde 2005).

Christensen, Siggaard & Veliyev (2023, “CSV”) ask whether machine learning (ML) beats HAR, and where any gain comes from. They run a 22-model horse-race — the HAR family (HAR, LogHAR, LevHAR, SHAR, HARQ), regularised regression (Ridge, Lasso, Elastic Net, adaptive- and post-Lasso), tree ensembles (bagging, random forest, gradient boosting) and feed-forward neural nets (four geometric-pyramid architectures, each a single net NN^1 and a 10-of-100 ensemble NN^{10}) — over two information sets: M_HAR (three RV lags) and M_ALL (plus nine macro/firm predictors). On 29 Dow Jones stocks, 2001–2017, they conclude ML beats HAR; the gain is small and class-specific on M_HAR but clear on M_ALL (Elastic Net $\approx -8\%$, random forest $\approx -10\%$, best NN $\approx -11\%$ at $h=1$), grows with horizon, and is led by random forests at one month ($\approx 40\%$ MSE reduction). Significance uses Diebold–Mariano (DM; 1995) and the Hansen–Lunde–Nason Model Confidence Set (MCS; 2011); interpretability uses Accumulated Local Effects (Apley & Zhu 2020).

We replicate the full methodology on the three stocks with available 5-minute data over 2016–2024 — seven years later than CSV, spanning the COVID shock and 2022 regime. AAPL and JPM are in CSV's cross-section (a check); AMZN joined the DJIA only in 2020, giving out-of-sample-stock evidence. We target a sound, transparent implementation and a critical comparison, not exact numbers.

2. Implementation and replication

Data. From 1-minute bars we keep the 09:30–16:00 session, apply a Barndorff-Nielsen et al. (2009)-style outlier filter, resample to 5-minute returns within each day (no overnight returns), and compute RV, signed semivariances and realised quarticity. The split is 70/10/20 chronological, as CSV. The daily lag RV_{t-1} predicts RV_t (a true one-step forecast).

Models. All 22 CSV models sit behind one interface. HAR is closed-form OLS; LogHAR carries the Jensen back-transform; HARQ uses the Bollerslev–Patton–Quaedlylieg (2016) insanity filter (clip to the in-sample range). Regularised models use a 100-point log- λ grid (CSV 1,000; the optimum is interior, so this is immaterial). Bagging/RF use Breiman–Cutler defaults; gradient boosting is validation-tuned. Following CSV (p.1691) any negative variance forecast is replaced by the in-sample minimum RV, applied uniformly.

Honest deviations. The NNs run on scikit-learn (PyTorch unavailable): ReLU not leaky-ReLU and no dropout, so early stopping is the regulariser in dropout's place. For tractability the regularised, tree and NN models are fixed-window; CSV roll the regularised models and trees and fix only the NNs — our fixed-window trees are the main deviation, exploited in §4. Implied volatility is proxied by VIX (CSV use licensed OptionMetrics). **Evaluation:** MSE (primary, as CSV) plus QLIKE (Patton 2011) as a noise-robust complement; DM with Newey–West HAC and the Harvey–Leybourne–Newbold correction; the HLN MCS (10,000 moving-block bootstraps) at $\alpha \in \{0.25, 0.10, 0.05\}$; ALE for variable importance. Thirty-seven unit tests cover the critical path.

3. Results

We select three results as most important — the M_ALL comparison at $h=1$, its behaviour across horizons, and the significance picture — because these are CSV's three headline claims.

h	Stock	LogHAR	EN	LA	RF	BG	NN2 ¹⁰	NN3 ¹⁰
1	AAPL	0.944	0.953	0.941	1.069	1.167	0.891	0.938
1	AMZN	1.000	0.970	0.881	0.963	1.140	0.879	0.861
1	JPM	0.924	0.953	0.951	0.952	1.054	0.912	0.950
22	AAPL	0.666	1.031	1.021	3.038	7.314	1.009	0.926
22	AMZN	1.214	1.085	0.901	3.885	6.719	1.129	1.245
22	JPM	0.630	1.012	0.976	5.274	13.835	0.860	0.716

Table 1. Out-of-sample MSE relative to HAR, M_ALL (ratio < 1 beats HAR). Full 22-model tables, DM stars and MCS sets in the Appendix.

At $h=1$ on M_ALL the best model beats HAR by ≈ 9 –14% on every stock (AAPL 10.9%, AMZN 13.9%, JPM 8.8%) — NN ensembles lead (0.86–0.91), Lasso close (0.88–0.95) — reproducing CSV's magnitude ($\approx 11\%$) and ordering (NNs best, regularisation helping, bagging weakest). On the apples-to-apples M_HAR set the gain shrinks to a few percent and only the NN ensembles beat HAR, with trees losing — again matching CSV.

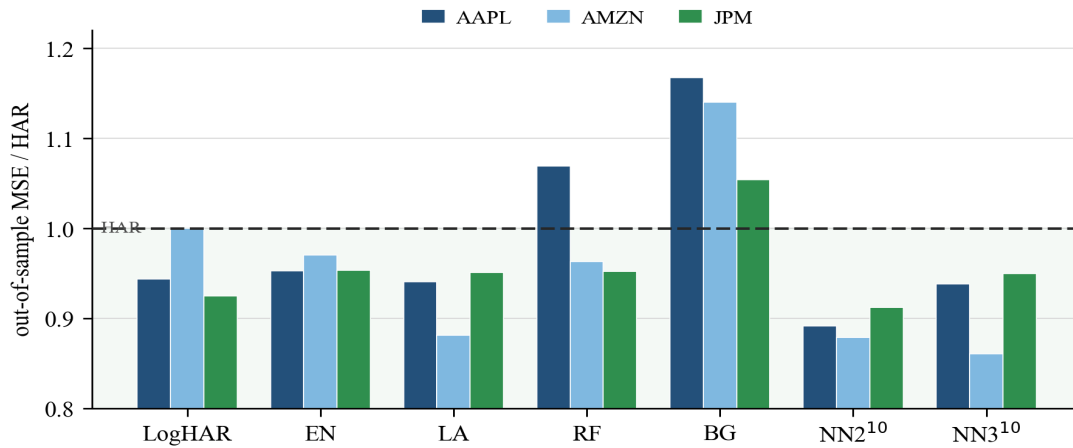


Figure 1. M_{ALL} MSE ratios vs HAR at $h=1$; bars below the dashed HAR line beat HAR.

Across horizons the tree picture diverges sharply and is our most informative result. CSV find ML's edge grows with horizon, RF leading at $h=22$; we find the opposite for fixed-window trees — at $h=22$ bagging is 7–14× HAR and RF 3–5× — while LogHAR dominates the HAR family (0.63–0.67 on AAPL/JPM) and the NN ensembles stay competitive. **Significance is thin:** at $h=1$ the MCS at 90% retains 20–21 of 22 models on every stock and never eliminates HAR; one-sided DM finds only 0–3 models per stock beat HAR at 5%, and under a Holm/BH multiple-testing correction almost nothing survives. Under QLIKE the ML advantage largely disappears.

4. Critical comparison and discussion

The replication matches CSV where the data allow (LogHAR strongest in the HAR family, NN ensembles best at $h=1$, regularisation helping on M_{ALL}), so the substantive contribution is the critique. Five points, each tied to a financial, econometric or ML reason; full development with page/table references in the Appendix.

(i) The “machine learning” headline is mostly information set and regularisation, not nonlinearity [ML]. CSV's own tables separate the effects their title bundles. On M_{HAR} (Table 2, $h=1$) only the NN ensembles beat HAR, by $\approx 5\%$; trees lose (bagging +15%) and regularisation ties (EN 0.999). The gain appears only on M_{ALL} , where plain OLS on the wider set (HAR-X) already earns 3.4%, Elastic Net 8.4% and the best NN 11.5% — regularisation does roughly half the work, nonlinearity the rest. Our path decomposition reproduces the split (regularisation +3–5pp, deep-NN +4–18pp, trees flat-to-negative). “Regularisation on a richer information set” would describe the result more honestly than “machine learning”.

(ii) CSV's long-horizon tree result depends on the daily re-fit, not the model class [econometric/financial]. CSV roll bagging/RF daily but fix the NNs once (p.1693, conceded “strictly disadvantageous to the NNs”), so “RF best at $h=22$ ” (Table 7, $\approx 40\%$ reduction) confounds *trees* with *re-estimated daily*. We isolate the two: with all ML fixed-window the trees collapse at $h=22$; refitting RF daily — CSV's scheme — roughly halves the AAPL ratio (3.04→1.55), pinning the daily re-fit, not “trees capture long memory”, as the load-bearing assumption (it does not fully reach CSV's $RF < HAR$, evidence the result is also sample-conditional). A model needing constant re-estimation to survive is the one that fails at a regime change, when forecasts matter most.

(iii) The significance machinery does not support the prose [econometric]. CSV's claim is a between-means statement (lower average MSE via DM). The MCS — the test of *distinguishable* accuracy — keeps HAR in the best set: their Figure 4 retains LogHAR at a high rate, and ours retains 20–21 of 22 models at $\alpha=0.10$ with HAR never eliminated. DM finds 0–3 models per stock beat HAR at 5%, and a multiple-testing correction (CSV apply none, despite ≈ 136 pairwise tests per stock) removes essentially every $h=1/h=5$ star. ML lowers average MSE a few percent but is not statistically distinguishable from HAR at the daily horizon.

(iv) The edge is loss- and regime-conditional [financial]. CSV evaluate on MSE only; under QLIKE (Patton's noise-robust loss) the advantage largely vanishes — HAR-X, Ridge, Elastic Net and the NNs are all QLIKE-worse than HAR on most stocks at $h=1$. MSE rewards fit on a few high-variance days; QLIKE weights the many ordinary days, where HAR is hard to beat. And on a 2016–2019-train / 2020–2024-test split (spanning the COVID crash and 2022) the edge collapses from $\approx 9\text{--}14\%$ to $\approx 0\text{--}2\%$ — no model beats HAR on JPM. Since proportional accuracy and turbulent regimes are what risk management cares about, the practical value of the edge is smaller than an MSE-only, calm-period comparison implies.

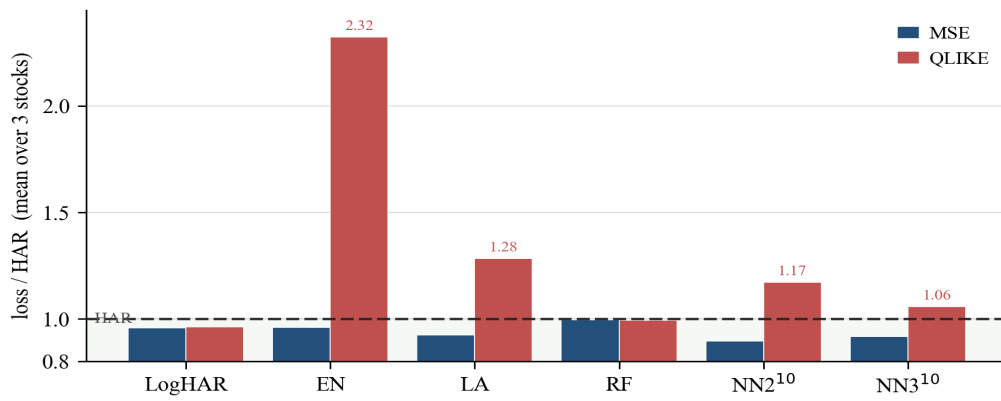


Figure 2. Models that beat HAR under MSE (left bars < 1) are mostly worse under the noise-robust QLIKE loss (right bars > 1); only LogHAR is robust to both. $h=1$, M_ALL , mean over three stocks.

(v) Long-memory interpretation and licensed-data dependence [econometric/reproducibility]. CSV credit “better approximating long memory”, but their evidence is *in-sample fitted* ACF (Figure 8), which cannot distinguish captured long memory from over-smoothing toward the unconditional mean — our *out-of-sample* ACF shows ML forecasts more persistent than realised RV, the over-smoothing signature. And the M_ALL gain leans on per-stock OptionMetrics implied volatility, the strongest extra predictor (Table A.5, $t=6.4$) yet licensed, unreproducible, and redundant once proxied by VIX (insignificant alongside IV, $t=0.09$). Where we match CSV the agreement is reassuring; where we differ, the difference is the contribution — enabled by a regime-spanning sample CSV could not observe.

Further critiques (full development in CRITIQUE.md). **(vi)** The comparison runs on a noisy proxy: RV estimates integrated variance with error and HARQ exists to correct it (BPQ 2016), yet the ML methods receive the same noisy regressors uncorrected — switching to the noise-robust realised kernel moves JPM $h=1$ MSE by $\approx 27\%$, larger than the entire ML-vs-HAR gap, so part of the “gain” is estimator noise, not model skill [econometric]. **(vii)** “Minimal tuning” is asymmetric: CSV search a 1,000-point λ grid and keep the best 10 of 100 NN seeds but leave RF at 2004 defaults — the methods that win are the most heavily searched [ML]. **(viii)** The 22-model space is narrow: every model’s out-of-sample errors correlate with HAR at $r \geq 0.79$ ($\approx 0.88\text{--}0.99$ for the models that beat HAR), so each winner is HAR plus a small orthogonal perturbation and its gain lives in under $\sim 20\%$ of error variance — which is also why combinations add little [ML]. **(ix)** “The basic HAR is superior to its offspring” (CSV p.1695) is split-dependent: their own Appendix A.1 flips it under a shorter window, so the headline ranking is conditional on the train/test design rather than structural [econometric].

References

Andersen & Bollerslev (1998, *Int. Econ. Rev.*); Apley & Zhu (2020, *JRSS-B*); Barndorff-Nielsen & Shephard (2002, *JRSS-B*); Barndorff-Nielsen, Hansen, Lunde & Shephard (2008, *Econometrica*); Bollerslev, Patton & Quaedvlieg (2016, *J. Econometrics*); Christensen, Siggaard & Veliyev (2023, *J. Financial Econometrics* 21(5), 1680–1727); Corsi (2009, *J. Financial Econometrics*); Diebold & Mariano (1995, *JBES*); Hansen & Lunde (2005, *J. Applied Econometrics*); Hansen, Lunde & Nason (2011, *Econometrica*); Patton (2011, *J. Econometrics*); Patton & Sheppard (2015, *Rev. Econ. Stat.*); Zou (2006, *JASA*).

Appendix (≤3 pages — full content in APPENDIX.md)

A1 full 22-model MSE tables (M_ALL h=1 & h=22); A2 the apples-to-apples M_HAR table; A3 critique evidence (path decomposition, fixed-vs-rolling tree experiment, MCS/DM significance, QLIKE reversal, realised-kernel, error correlation); A4 ALE variable importance; A5 code excerpts; A6 data construction and disclosed deviations. Set the wide A1 tables landscape (as CSV do). Machine-readable tables in `outputs/`; full critique in `CRITIQUE.md`.